

Excel Structure

Lesson 24 — Workbook, Worksheet and Cell Anatomy

JKSSB Computer Awareness

October 8, 2026

Excel and the Workbook

- ▶ **Microsoft Excel** is a spreadsheet program that stores, calculates, and analyses data in a grid.
- ▶ A **workbook** is the Excel **file** that you save and open.
- ▶ One workbook can hold **many worksheets**, also called sheets.
- ▶ Worksheet **tabs** at the bottom of the window let you move between them.

The one idea

A workbook is the file; a worksheet is a single page inside that file.

Workbook vs Worksheet

Point	Workbook	Worksheet
What it is	The Excel file	One sheet inside a workbook
How many	One file	Many per workbook
Saved as	A single file name	Part of that file
Tab	—	Tab at the bottom

Trap alert

A workbook is **not** the same as a worksheet. The worksheet is a part of the workbook, not the whole file.

The Worksheet Grid: Rows and Columns

- ▶ Every worksheet is a grid of **rows** and **columns**.
- ▶ **Rows** run horizontally and are named by **numbers** on the left (1, 2, 3, ...).
- ▶ **Columns** run vertically and are named by **letters** across the top (A, B, C, ...).
- ▶ Column labels go A–Z, then AA–AZ, then BA, and so on up to the last column.

Read the row number on the left edge and the column letter on the top edge of the sheet.

Grid Dimensions (Direct Recall)

Property	Value in Excel 2007 and later
Total rows	1,048,576 (numbered 1 to 1,048,576)
Total columns	16,384 (labelled A to XFD)
First column	A
Last column	XFD

Remember the two numbers

Rows: **1,048,576**. Columns: **16,384**. The last column is **XFD**.

The Cell

- ▶ A **cell** is the **intersection** of one row and one column.
- ▶ It is the basic box in which you type data.
- ▶ Every cell has a unique address, formed from its column letter and its row number.
- ▶ A worksheet is simply a very large collection of these boxes.

Definition

A cell is the small rectangular box formed where a row and a column meet.

Cell Address (Cell Reference)

- ▶ A **cell address** names one specific cell.
- ▶ It is written as the **column letter first**, then the **row number**.
- ▶ Example: **B5** means column B, row 5.
- ▶ Example: **C7** means column C, row 7.

Worked example

The cell at column D and row 12 is written **D12** — letter first, then number. Writing 12D or D:12 is wrong.

The Active Cell

- ▶ The **active cell** is the cell that is currently **selected**.
- ▶ It is surrounded by a thick border.
- ▶ Whatever you type is entered into the active cell.
- ▶ Its address appears in the **Name Box**, and its contents appear in the **Formula Bar**.

Trap alert

Only **one** cell is active at a time, even when a whole range of cells is selected.

The Formula Bar

- ▶ The **Formula Bar** is the long white bar **below the Ribbon** and above the grid.
- ▶ It **shows the contents** of the active cell.
- ▶ For a formula, it shows the **formula itself**, not the result shown in the cell.
- ▶ You can also **edit** the active cell's contents directly in the Formula Bar.

Keep it straight

The cell shows the **result**; the Formula Bar shows the **formula or data** behind it.

The Name Box

- ▶ The **Name Box** is the small box at the **left end of the Formula Bar**.
- ▶ It normally shows the **address of the active cell**.
- ▶ If a range is selected, it shows the **name or address of that range**.
- ▶ You can type an address into the Name Box to jump straight to that cell.

Range

- ▶ A **range** is a group of two or more cells treated as one block.
- ▶ It is written with the two **corner addresses** separated by a colon (:).
- ▶ Example: **A1:C10** means columns A to C, rows 1 to 10.
- ▶ A range is usually **rectangular** and its cells are contiguous.

Worked example

The range B2:D5 covers 3 columns (B, C, D) and 4 rows (2–5), so it contains $3 \times 4 = 12$ cells.

Cell vs Range (Near-Neighbour)

Point	Cell	Range
What it is	One box	A block of cells
Written as	B5	B2:D5
Number of cells	Exactly one	Two or more

Trap alert

B5 is a single cell, not a range. A range needs **two** corner addresses joined by a colon, such as **B5:D9**.

Basic Data Types

- ▶ **Text** (also called a label): words, names, headings.
- ▶ **Number**: values used in calculations.
- ▶ **Date and time**: stored as numbers but shown as dates.
- ▶ **Formula**: an entry beginning with **=** that calculates a result.
- ▶ Excel also accepts **TRUE/FALSE** (logical values) and shows errors such as **#VALUE!**.

Text vs Numbers: Default Alignment

Data type	Default alignment	Example
Text	Left	Roll No
Number	Right	4500
Date	Right	15-08-2026

Quick check

If a value sits on the **right** of the cell, Excel has treated it as a **number**; if it sits on the **left**, it is **text**.

Moving Around a Worksheet

- ▶ Click any cell to make it the **active cell**.
- ▶ Use the **arrow keys** or **Tab** / **Enter** to move.
- ▶ The worksheet **tabs** switch between sheets in a workbook.
- ▶ The **Name Box** jumps to a typed address; the Formula Bar reads and edits the active cell.

These four elements — grid, Name Box, Formula Bar, tabs — are the basic navigation anatomy of Excel.

Trap Alert: Facts to Keep Straight

- ▶ A **workbook** is the file; a **worksheet** is one sheet inside it.
- ▶ A **cell** is one box; a **range** is two or more cells.
- ▶ Cell address = **column letter, then row number** (B5).
- ▶ Total rows: **1,048,576**; total columns: **16,384**.
- ▶ Last column: **XFD**.
- ▶ **Name Box** shows the address; **Formula Bar** shows the contents.

Lesson 24: Recall

- ▶ A workbook is the Excel file; it can contain many worksheets.
- ▶ A worksheet is a grid of numbered rows and lettered columns.
- ▶ Excel 2007+ has 1,048,576 rows and 16,384 columns (A to XFD).
- ▶ A cell is the intersection of a row and a column; the active cell is the one selected.
- ▶ A cell address is column letter plus row number (B5); a range is a block such as A1:C10.
- ▶ The Name Box shows the address; the Formula Bar shows the contents.
- ▶ Basic data types: text, number, date/time, and formula.